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Problem Statements - General Instructions

Competition: 26th September, 10:00 AM → 27th September, 10:00 AM

Problem Statement Release: 26th September, 10:00 AM

Form Submission [link](#) → Stops taking responses on 27th September 10:00 AM

Problem Statements

There are three categories; choose any 1 category:

[Web Development](#) - 3 problem statements

[AI/ML](#) - 3 problem statements

[Systems & Cybersecurity](#) - 3 challenges

Web Development and AI/ML teams must select **one problem statement** out of 3 and work on it throughout the competition.

Systems & Cybersecurity teams will receive a virtual environment containing three challenges. [Go through the guide.](#) Teams may attempt the challenges according to the instructions provided.

Do not split your team across multiple problem statements. Choose 1 carefully and use the 24-hour window effectively. We insist to discuss the first 30 mins, among yourselves which category and which problem statement the team should solve. All Problem statements are made in such it can be built in 24 hrs

Development

Build a complete solution that directly addresses the requirements of the selected problem statement.

Keep your project **clean, readable, modular, and maintainable**. Your repository should clearly reflect the development work carried out by your team, with meaningful commits and contributions from team members.

Documentation

Web Development & AI/ML: Submit a **5–6 page structured project document** covering the project overview, architecture, approach, methodology, implementation, technologies used, and relevant technical decisions.

Systems & Cybersecurity: Submit a **complete solution write-up** covering your approach, reasoning, commands and tools used, observations, methodology, and how the challenges were solved, reflecting your thought process.

AI Coding Tools

AI coding tools are allowed. If AI coding agents are used, retain the relevant instruction/configuration files in your repository, such as **CLAUDE .md**, **AGENTS .md**, **GEMINI .md**, **.cursorrules**, **SKILLS.md**, or equivalent files.

Final Submission

Web Development: GitHub Repository, Project Documentation, Demo Video, AI Agent Files (if applicable), and **Deployed URL**.

AI/ML: GitHub Repository, Project Documentation, Demo Video, and AI Agent Files (if applicable).

Systems & Cybersecurity: Solution Write-up.

Ensure all submitted links are accessible and your repository contains the complete work.

Evaluation

Evaluation will consider the **quality and completeness of the solution, relevance to the problem statement, technical implementation, repository organisation, development practices, documentation, and the team's approach to solving the problem.**

For Systems & Cybersecurity, the **challenges solved and the quality of the documented methodology and reasoning** will also be considered.

Note: Submission time may be considered as a tie-breaker where necessary.

Submission Deadline

The competition and submission window close at **10:00 AM on 27th September**.

The submission form will stop accepting responses **exactly at 10:00 AM**. Submit early and verify all links before submitting.

Only the Team Leader should submit the form. Duplicate submissions will not be considered.

Make the most of the 24 hours. Read carefully, plan early, build smart, and document your work.

Web Development

Note: Choose any 1 problem statement only

Across all three, "hero" represents security services/personnel — the humans or systems who actually respond when something goes wrong.

1. Spider-Sense: Live Threat Map

A web platform that tries to feel danger before it fully arrives — surfacing incidents and risk signals across a city as they emerge, with the experience itself growing more alert as something gets closer to happening. The idea is less about a static dashboard and more about a interface that feels alive to what's unfolding around it, nudging security services to react before it's too late.

2. Daily Bugle News Engine

A platform for citizens to report suspicious activity or incidents, where the real challenge is separating credible signal from noise, rumor, and outright fake reports — much like the Bugle separating real heroics from tabloid nonsense. What eventually gets treated as "verified" shapes what security services actually act on, so the trust layer matters as much as the reporting itself.

3. Web-Shooter Dispatch

A system built around one core question: when something goes wrong, how does the right hero end up there fastest? It's about connecting incidents to available security services in real time, coordinating who responds, and making sure nothing falls through the cracks — even when multiple people or teams could reasonably respond to the same thing.

Artificial Intelligence / Machine Learning

Note: Choose any 1 problem statement only

1. "The Daily Bugle Truth Serum" - Fake Text Detection

The Web We Weave: J. Jonah Jameson's headlines were always loud, but lately they're not even written by people. The villains have flooded New York with AI-generated news articles, fake Spider-Man "interviews," and fabricated social posts, all crafted to turn the city against its hero. It reads real. It sounds real. But it's machine-made poison, and it spreads faster than Peter can swing. New Yorkers need a way to tell whether the words on their screen came from a person or a machine.

Your Mission: Give the city a "Spider-Sense" for AI-written text, a way to know whether an article, post, or message was written by a human or generated by a machine, before they believe it and pass it on.

2. "Karen's Ear" - AI Emergency Dispatch Assistant

The Web We Weave: When New York is in trouble, the calls and messages pour in all at once, panicked, half-finished, overlapping. "There's smoke on 5th." "Someone's stuck under the rubble." "The lizard-thing is heading downtown." Buried in that flood of words is exactly the information Spider-Man needs: what's happening, where, and how urgent. But no human can read it all fast enough, and by the time the important message is spotted, it's too late. Karen, Peter's AI assistant, needs to cut through the chaos and tell him where to swing first.

Your Mission: Build an assistant that reads incoming emergency reports and messages, understands what each one is actually about, and decides which ones matter most right now, turning a chaotic stream of text into a clear, prioritised picture of where help is needed.

3. "Spider AI: The Silent Guardian" - Ambient SOS

The Web We Weave: Not everyone has Spider-Man nearby when danger hits. The scream in an empty parking garage, the fall no one witnesses, the quiet "leave me alone" that no one hears—these moments pass without anyone coming to help, simply because no one knew they were happening. Peter's Spider-Sense warns him the instant someone's in trouble. Ordinary people have no such gift.

Your Mission: Give everyone their own Spider-Sense, a quiet guardian that recognises when someone is in genuine danger and makes sure a call for help reaches the people who can answer it.

Systems & CyberSecurity 🐉 well u came here !

Note: Here, the more no. of challenges you solve the more number points you get

Read the [CTF participation guide](#) where you will solve these

Challenge 1 - Dehashing the Goblin's Secret

A suspicious string has been hidden behind a hashing mechanism. Your task is to **write a dehashing algorithm** capable of recovering the original string from the given hash.

Use the provided challenge environment and investigate the files carefully to determine the hashing method and recover the required string.

Read the instructions provided inside the Challenge 1 directory before beginning. They contain the information and clues necessary to understand the challenge.

Objective: Recover the particular string required by the challenge and submit it using the provided submission mechanism.

Challenge 2 - The Daily Bugle Archive

The Daily Bugle has left behind an executable containing a hidden encoded string. Your task is to investigate the executable using **UNIX-compliant tools** and uncover the hidden information. You will need to inspect the executable and identify strings or encoded data that are not immediately visible from normal execution.

Read the instructions provided inside the Challenge 2 directory before beginning. They contain the clues and direction needed to approach the executable.

Objective: Find and decode the required string hidden within the executable, then submit the recovered answer using the provided submission mechanism.

Challenge 3: Web-Slinger's Secret: Category Focus

Category Focus: Steganography · Digital Forensics · Cryptography · File Analysis

What looks like an ordinary PNG image conceals a multi-layered secret beneath its surface.

Your objective is to **investigate the supplied image, uncover the hidden data, and reconstruct the final flag**. The image contains deliberate anomalies that prevent a simple image viewer from revealing everything. Solving the challenge will require careful analysis of the file and the information hidden within it.

Task Overview

You will need to:

- Investigate the supplied PNG beyond what a normal image viewer reveals.
- Identify and repair a deliberate issue in the file.
- Extract hidden information from the image.
- Follow multiple layers of encoding and transformation.

- Use information derived from the image itself to recover the final data.
- Decode the final representation to obtain the flag.

This is a **multi-stage investigation**. Each discovery provides information required for the next stage.



Environment

You will be provided with:

- A PNG image associated with the challenge.
- Access to a Linux environment through SSH.
- The challenge instructions and necessary environment tools.

Read the instructions inside the Challenge 3 directory before beginning. They contain important details and hints for progressing through the investigation. You are encouraged to inspect the file using command-line tools and write your own scripts when necessary.

Automation and assistance tools, including custom scripts and LLMs/code generators, are permitted.

Write-ups MUST include detailed logs and documentation of the commands, scripts, observations, and methodology used to solve the challenge.

Hint: If a tool produces unexpected output or the image appears corrupted, don't immediately assume the challenge is broken. Recheck your previous steps and investigate the file more closely.

CTF Participation Guide

1. Connect to the CTF

Use SSH from Windows Terminal, PowerShell, Linux, or macOS:

None

```
ssh ctf@16.112.218.133
```

For the first time, it will ask for permission; type **yes**: then the **password** is : 123



Then you will first enter your team lead's credentials:

None

```
Team name:
```

```
Leader email:
```

After team authentication, enter your own credentials (even if you are the lead) :

None

Member name:

Member email:

You will then be placed inside your team's workspace.

2. Explore the Workspace

Your workspace is:

None

```
cd /workspace
```

Check the available challenges:

None

```
ls
```

```
tree
```

Challenges are independent and can be solved in any order.

```
@team-UIH578Y9:/workspace$ tree
.
|-- game
|   |-- Challenge1
|   |   |-- instructions
|   |   |   |-- challenge1.md
|   |   |-- journals
|   |   |   |-- aunt-may.txt
|   |   |   |-- harry-osborn.txt
|   |   |   |-- otto-octavious.txt
|   |   |-- peter-parker-dairy.txt
|   |   |-- secrets
|   |   |   |-- laundry.txt
|   |   |   |-- osborn-corp.txt
|   |   |   |-- pizza-parlour.txt
|   |-- Challenge2
|   |   |-- bugle
|   |   |-- instructions
|   |   |   |-- challenge2.md
|   |-- Challenge3
|   |   |-- challenge.png
|   |   |-- instructions
|   |   |   |-- challenge3.md
|-- submit
```

3. Read the Instructions

Each challenge contains its own instructions. **Read them before starting. (cat or vim)**

Shell

```
cat /workspace/game/Challenge3/instructions/challenge3.md
```

4. Submission

Use the provided submission program:

None

```
./submit <challenge-number> "<answer>"
```

Example:

None

```
./submit 1 "your_answer"  
./submit 2 "your_answer"  
./submit 3 "your_answer"
```

The system will tell you whether your submission is correct.

5. Tools Provided

The environment already contains the tools required to solve the intended challenges.

Some useful tools:

None

Linux utilities

```
ls · cd · find · tree · grep · strings · file
```

Binary analysis

```
xxd · objdump · readelf
```

Image / forensics
pngcheck · exiftool

Encoding / hashing
base64 · sha256sum

Programming
Python 3 · Pillow · GCC · G++ · Make

You are encouraged to write scripts when useful.

You do not need to install additional tools on the CTF server.

You read till the last awww..!! That's so sweet of you 🥰